

Marios Vardalachakis

AI Engineer | Heraklion, Greece

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SUMMARY

Practical experience in designing and developing industrial-strength machine learning systems in the Horizon Europe research context as an AI/ML software developer. Strong expertise in end-to-end AI delivery with Python and R programming languages, PyTorch/TensorFlow deep learning frameworks, PostgreSQL databases, and REST APIs, as well as experience in evaluation pipelines, privacy-utility tradeoffs, and JavaScript-based data visualization for designing interactive analytics dashboards and research demonstrators. Research work for an M.Phil. degree involved anonymization techniques in Machine Learning/DL, including k-anonymity, autoencoder-based anonymization of IoT time-series data, Differential Privacy, Federated Learning, and GAN-based synthetic data generation.

SKILLS

- **AI/ML and Data:** Python, Scikit-learn, PyTorch, TensorFlow, Data Preprocessing, Model Evaluation, Differential Privacy
- **Backend/Apis/&MLOps:** FastAPI, Flask, R (plumber), PostgreSQL, Docker, CI/CD (GitHub Actions), Linux
- **Software Engineering:** Git, Clean Architecture, Secure Coding, Privacy-by-Design
- **Visualization/Prototyping:** R (Shiny), JavaScript, Plotly, D3.js, dc.js, Shiny Dashboards

EXPERIENCE

AI/ML Full Stack Engineer | M. Phil Researcher (Intelligent Systems)

PRIVACT P.C. & PrivaSi Lab – University of the Aegean |

2024 – Present

Project: InvisseeAI – Advanced Healthcare Data Anonymization Platform

 GitHub

- Designed and implemented InvisseeAI as an end-to-end AI-driven platform for healthcare data anonymization and privacy-preserving analytics.
- Developed backend RESTful APIs using R (Plumber) to expose anonymization workflows and AI pipelines as production-ready services.
- Implemented classical anonymization techniques (k-anonymity, l-diversity, t-closeness) and secure multiparty computation (SMPC) methods to enable privacy-preserving data sharing and processing.
- Built secure data processing pipelines aligned with GDPR, privacy-by-design, and data minimization principles.
- Applied Federated Learning and Differential Privacy to ensure privacy guarantees with minimal utility loss.
- Designed modular, layered software architecture following clean code principles for scalability and maintainability.
- Contributed to experimentation, testing, and technical validation in an R&D research environment.
- Supported EU Horizon Europe activities through deliverables and demonstrator development (SECURED project).

R&D Engineer (Internship & BSc Thesis)

Computational Biomedicine Lab - ICS-FORTH

2016 – 2018

Project: ShinyAnonymizer: A tool For Anonymizing HealthCare Data

 GitHub

- Developed ShinyAnonymizer, an interactive platform enabling anonymization of sensitive healthcare datasets.
- Built real-time dashboards for dataset management, privacy-parameter configuration, and visualization.
- Integrated anonymization, hashing, and encryption techniques with PostgreSQL backend components.
- Contributed to Horizon 2020 iManageCancer research demonstrator development.

Additional projects : mariosvard.github.io/mariosvard

EDUCATION

M.Phil. by research – Intelligent Systems(Fully Remote)

2024-2026

University of the Aegean

M.Sc. – Software Engineering(Fully Remote)

2020 -2024

Hellenic Mediterranean University

B.Sc. – Informatics Engineering

2011-2020

Hellenic Mediterranean University

THESIS RESEARCH PROJECT (M.Phil)

Examining the Role of Machine Learning & Deep Learning in Anonymization Techniques

- Analyzed healthcare datasets using the traditional anonymization techniques (k-anonymity, l-diversity, and t-closeness).
- Developed synthetic data generating pipelines using GANs for data sharing in fraud detection.
- Federated learning and applied differential privacy ensure security for privacy with minimal loss of usefulness.
- Convolutional Autoencoder models were developed to preserve temporal utility while anonymizing IoT time-series data.

Publications

InvisseeAI: Advanced Healthcare Data Anonymization Platform, IISA 2025 ([DOI](#)).
ShinyAnonymizer Enhanced Version and Beyond, Applied Sciences, 2024 ([DOI](#))

For More Publications: [IEEE](#)